Effect of different curcumin dosages on human gall bladder.

Our previous study demonstrated that curcumin, an active compound of Curcuma xanthorrhiza and C. domestica, produces a positive cholekinetic effect. A 20 mg amount of curcumin is capable of contracting the gall bladder by up to 29% within an observation time of 2 h. The aim of the current study was to define the dosage of curcumin capable of producing a 50% contraction of the gall bladder, and to determine if there is a linear relationship between doubling the curcumin dosage and the doubling of gall bladder contraction. A randomised, single-blind, three-phase, crossover-designed examination was carried out on 12 healthy volunteers. Ultrasonography was carried out serially to measure the gall bladder volume. The data obtained was analysed by analysis of variance (ANOVA). The fasting volumes of gall bladders were similar ($P > 0.50$), with 17.28 +/- 5.47 mL for 20 mg curcumin, 18.34 +/- 3.75 mL for 40 mg and 18.24 +/- 3.72 mL for 80 mg. The percentage decrease in gall bladder volume 2 h after administration of 20, 40 and 80 mg was 34.10 +/- 10.16, 51.15 +/- 8.08 and 72.25 +/- 8.22, respectively, which was significantly different ($P < 0.01$). On the basis of the present findings, it appears that the dosage of cucumin capable of producing a 50% contraction of the bladder was 40 mg. This study did not show any linear relationship between doubling curcumin dosage and the doubling of gall bladder contraction.